

Max. Marks: 20

EC1002-1 – APPLIED DIGITAL LOGIC DESIGN

Max. Marks: 20

Unit – I

		Unit – I	Marks	BT*	CO*	PO*
1.	a)	Show the implementation of a full adder using two half adders and OR gate with relevant equations.	6	L*2	3	1
	b)	Implement the following functions using a 4-to-16-line decoder.				
	i.	$f_1(w, x, y, z) = \bar{w}xy + xyz + \bar{w}xy\bar{z}$				
	ii.	$f_2(w, x, y, z) = \bar{w}xyz + yz + wxy\bar{z}$	4	L3	3	1
2.	a)	With a neat diagram and relevant example explain the working of 4-bit parallel adder/subtractor.	6	L2	3	1
	b)	Implement the function $f(x, y, z) = \prod M(0, 2, 4, 5)$ using				
	i.	8:1 Mux				
	ii.	4:1 Mux by considering x and z as select lines	4	L3	3	1

3.	a)	With function table and timing diagram explain the working of Positive edge triggered D Flip-Flop.	6	L2	4	1
	b)	Explain one-bit comparator using truth table and logic diagram. Also, implement a 1-bit comparator using a 2-to-4-line decoder.	4	L3	3	1
4.	a)	Explain the operation of Master-Slave JK flip-flop using NAND only with reference to the clock signal.	6	L2	4	1
	b)	With the help of the function table, explain the working of the SR flip-flop and show how SR flip-flop can be modified to D flip-flop.	4	L3	4	1
